

Practical 1

28 Jan 2026

Questions

Instructions

1. Upload your answers (along with codes) in the moodle with naming “roll no._p1”.
2. Do not copy codes, it will be checked for plagiarism

1. Download the EMBOSS package (<http://emboss.sourceforge.net/download/>) and copy to your Windows system. In case of **Linux** use the command: **sudo apt-get install jemboss**. For Mac users, use the online tool links given in the instructions doc.

2. Using EMBOSS, find the complementary strand for the sequence:

CTCGGATTTGTAAAGATCATGATCTCATACATAGTACCTAGCCATTG

3. Write a program to find the complementary strand for the sequence in given in Q2.

4. Using EMBOSS,

(i) find the protein sequence for

GACATTGTGAACAGTAAAAAAGTCCATGCAATGCGCAAGGAGCAGAAGAGGAAGCAGGGCAAGCAGCGCTCCATGGGC
TCTCCCATGGACTACTCTCCTCTGCCCATCGACAAGCATGAGCCTGAATTTGGTCCATGCAGAAGAAACTGGATGGG

(ii) Identify the reading frame equivalent to the following sequence.

PIQFSSAWTKFRLMLVDGQRRVVHGRAHGALLALLPLLLLAHCMDFFTVHNV

5. Write a program to translate the given DNA sequence (refer Q4) to the protein sequence.

6. Write a code to search for the following strings ‘AAG’, ‘GTC’, ‘GAG’, ‘ACTA’, and ‘ATAT’ in the DNA sequence provided in Q4. The program should print the total number of matches for the each of the given strings and the start positions of the matches

Example: Enter the string: ACTA
Total match:1
Position of match: 88

7. Familiarize with other applications in EMBOSS. For example, melting temperature, bending, curvature etc.

8. Write a program to compute the average base stacking energy for the sequence given in Q2 (AA: -4; AT: -7; AC: -5; AG: -11; TA: -7; TT: -2; TC: -3; TG: -4; CA: -9; CT: -5; CC: -6; CG: -7; GA: -9; GT: -6; GC: -4; GG: -11).

9. Compute the average melting temperature of the following sequences using Seq2Feature tool (<https://www.iitm.ac.in/bioinfo/SBFE/index.html>) and comment on the results (Enter one sequence at a time in fasta format)

(i) ATATATATAT

ii) GCGCGCGCGC

10. Calculate the AT and GC content of the sequence AAATGGCCCTAA using Seq2Feature tool

Deadline: 03-02-2026